MODULE 3: PROGRAMMING FOR BIOLOGICAL DATA

Session 8: Laboratory Quality Control

Levey-Jennings Charts & Westgard Rules

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Learning Objectives

Process Control

Understand Statistical Process Control (SPC) in the clinical lab.

Visualizations

Create professional Levey-Jennings charts using R.

Westgard Rules

Implement rules for QC rejection and warning.

Error Detection

Detect systematic shifts, trends, and random errors.

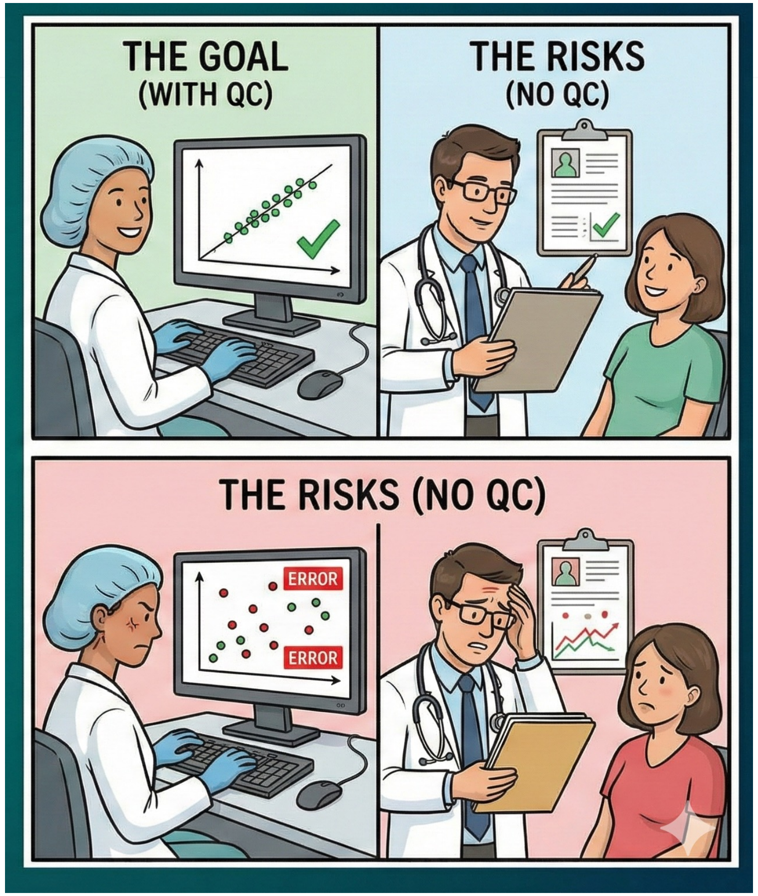
Why Quality Control Matters

The Goal

Ensure patient results are accurate, precise, and reliable.

The Risks of No QC

- ✕Calibration drift goes unnoticed.
- ✕Reagent degradation affects results.
- ✕Patient care decisions based on incorrect data.



Part 1: Statistical Process Control

Foundations of Laboratory Quality

The QC Workflow

Run QC Sample (Known Value)

↓

Compare Measured vs. Expected

↓

Within Acceptable Limits?

YES
Release Results

NO
STOP & Investigate

Control Materials

What are they?

- Samples with known/assigned values.
- Run alongside patient samples.
- Available in multiple levels (Low, Normal, High) to cover the analytical range.

Where do they come from?

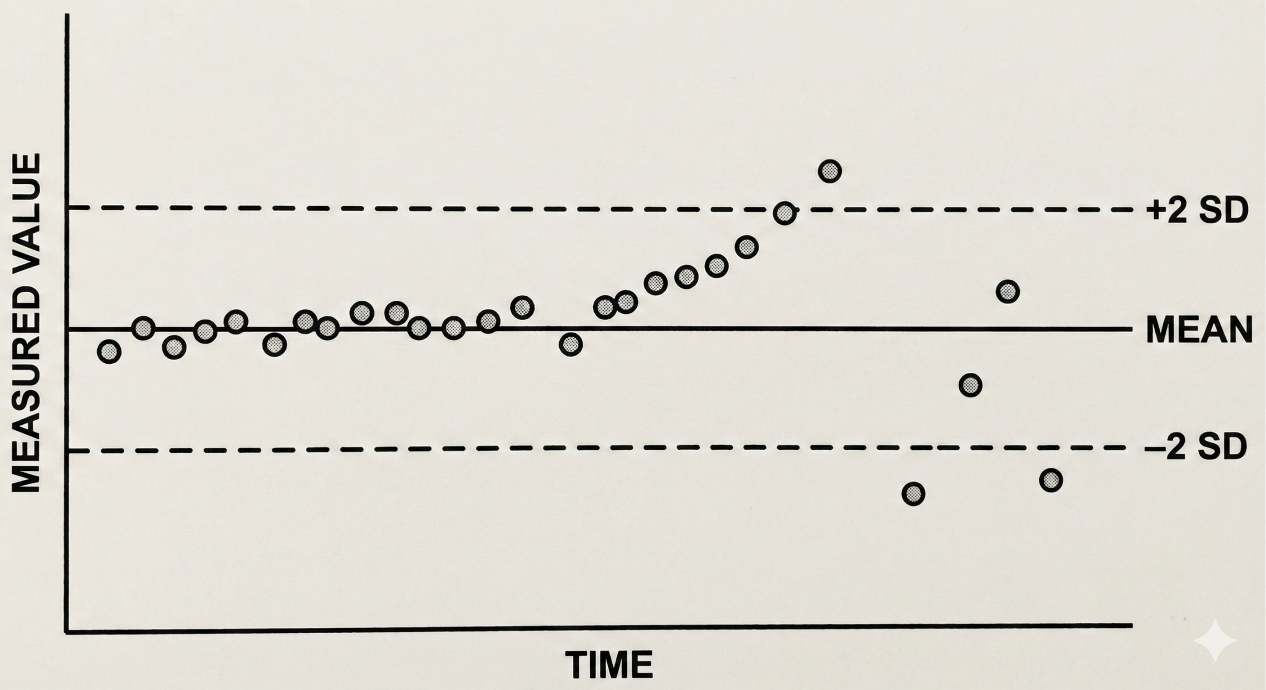
- Commercial manufacturers (e.g., Bio-Rad, Roche).
- Assigned values based on large peer group data.
- Mimic human specimens (matrix effect).

The Levey-Jennings Chart

Origin: Adapted from Shewhart industrial control charts for the clinical laboratory in 1950.

Purpose: To visualize QC performance over time, making it easy to spot trends, shifts, and random errors.

It places time on the X-axis and the measured value on the Y-axis against statistical limits.



Structure of an LJ Chart

Component	Description
X-axis	Time (days, runs, batches)
Y-axis	QC measured value
Center Line	Mean (Target Value)
Warning Limits	±2 Standard Deviations (SD)
Control Limits	±3 Standard Deviations (SD)

Based on Normal Distribution

QC limits are based on Gaussian statistics:

- **±1 SD:** 68.3% of values
- **±2 SD:** 95.4% (1 in 20 outside)
- **±3 SD:** 99.7% (1 in 370 outside)

If a value exceeds ±3SD, it is statistically highly unlikely to be due to chance alone.

Establishing Baseline Statistics

Rule: Use the first 20 data points to establish a reliable mean and SD.

```
# Assume 'qc_data' has a 'Value' column
# Extract first 20 QC values
baseline <- qc_data$Value[1:20]

# Calculate baseline statistics
baseline_mean <- mean(baseline)
baseline_sd <- sd(baseline)

# Note: These values define your chart limits!
```

Calculating Control Limits

```
# Control limits upper_1sd <- baseline_mean + 1 * baseline_sd lower_1sd <-
baseline_mean - 1 * baseline_sd upper_2sd <- baseline_mean + 2 * baseline_sd
lower_2sd <- baseline_mean - 2 * baseline_sd upper_3sd <- baseline_mean + 3 *
baseline_sd lower_3sd <- baseline_mean - 3 * baseline_sd
```

Creating a Basic LJ Chart (Base R)

```
plot(qc_data$Day, qc_data$Value,
     type = "b", pch = 19,
     main = "Levey-Jennings Chart",
     xlab = "Day", ylab = "QC Value")

# Add control lines
abline(h = baseline_mean, col = "green")
abline(h = upper_3sd, col = "red")
abline(h = lower_3sd, col = "red")
```

Base R provides a quick way to visualize the data, but for reports and dashboards, ggplot2 is preferred.

Professional Chart with ggplot2

ggplot(qc_data, aes(x = Day, y = Value)) + # Mean line geom_hline(yintercept = baseline_mean, color = "darkgreen") + # +/- 2SD lines geom_hline(yintercept = c(upper_2sd, lower_2sd), color = "orange", linetype = "dashed") + # +/- 3SD lines geom_hline(yintercept = c(upper_3sd, lower_3sd), color = "red") + # Data geom_line(color = "steelblue") + geom_point(size = 3, color = "steelblue")

Color Coded Control Zones

Zone	Range	Color	Interpretation
Green	±1 SD	Safe	Normal variation
Yellow	±1-2 SD	Warning	Watch closely
Orange	±2-3 SD	Alert	Investigate (Systematic?)
Red	> ±3 SD	STOP	Do not release results

Adding Colored Bands

```
ggplot(qc_data, aes(x = Day, y = Value)) +
  # Green zone (+/- 1SD)
  geom_rect(aes(xmin = -Inf, xmax = Inf,
                ymin = lower_1sd, ymax = upper_1sd),
            fill = "lightgreen", alpha = 0.3) +
  # Yellow zone (+/- 1-2SD)
  geom_rect(aes(xmin = -Inf, xmax = Inf,
                ymin = upper_1sd, ymax = upper_2sd),
            fill = "lightyellow", alpha = 0.3)
  # ... continue for other zones
```

Using geom_rect creates background zones that make visual interpretation instant.

Part 2: Westgard Rules

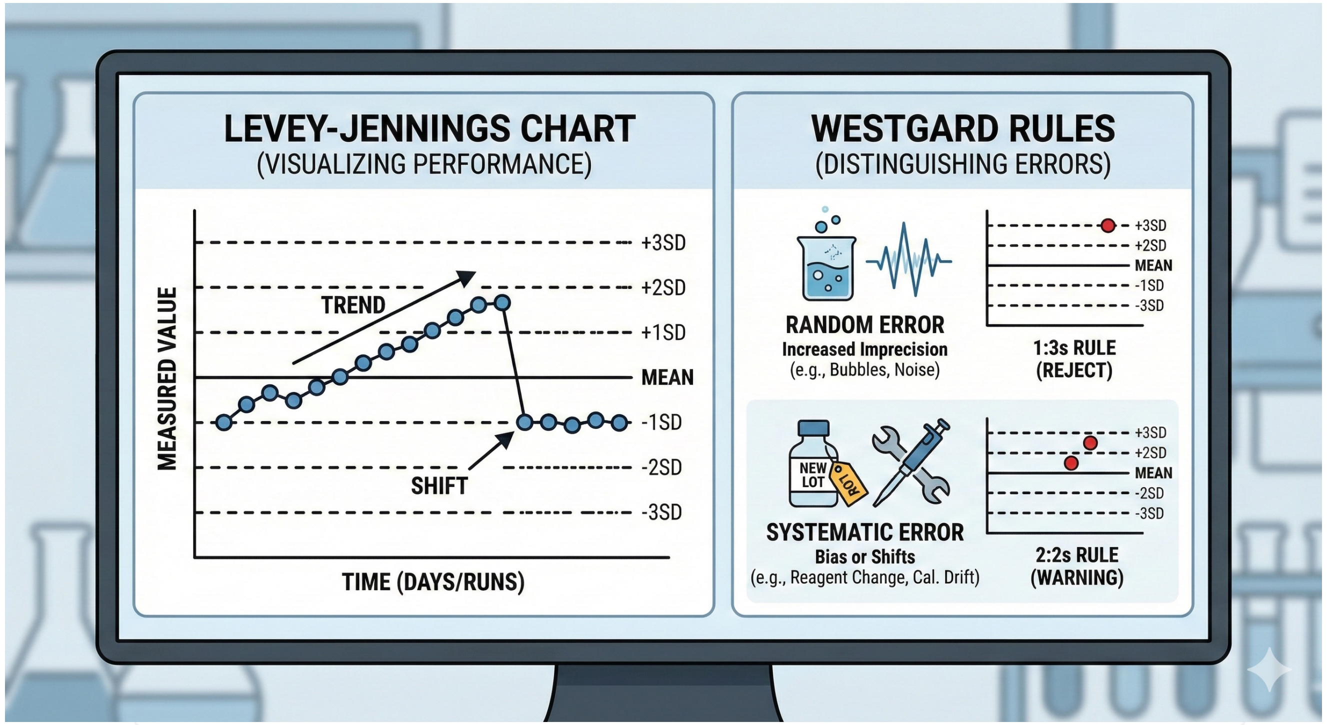
Advanced Error Detection

Why Westgard Rules?

Simple "outside ±3SD" checks are not enough!

James Westgard developed a multi-rule system to distinguish between:

- **Random Errors:** Increased imprecision (bubbles, electrical noise).
- **Systematic Errors:** Bias or Shifts (reagent lot change, calibration drift).



The 1:3s Rule

Definition

One QC value exceeds ±3 SD.

Action: REJECT

This is usually due to **Random Error**.

The run must be rejected and QC repeated.

Implementation

```
# Check 1:3s
violation_1_3s <- abs(value - mean) > 3
* sd
```

1:3s Statistics

0.27%

Probability of exceeding ±3SD by chance alone.

Low False Rejection

Because it is so rare, it is a high-confidence rule.

Rejection Rule

If triggered, assume something is wrong.

The 2:2s Rule

Definition

Two **consecutive** QC values exceed ±2 SD **on the same side** of the mean.

Error Type

Systematic Error. This often indicates a shift in the mean has occurred.

Action

Warning / Investigate.

Implementing 2:2s Rule

Logic for loop # Check if current AND previous > +2SD both_above <- values[i] > mean + 2*sd && values[i-1] > mean + 2*sd # Check if current AND previous < -2SD both_below <- values[i] < mean - 2*sd && values[i-1] < mean - 2*sd violation_2_2s <- both_above || both_below

The R:4s Rule

Definition

The **Range** between two consecutive values exceeds 4 SD.

Example: One point at +2.1 SD, the next at -2.1 SD. (Distance = 4.2 SD).

Implication

Random Error.

Indicates high imprecision in the system.

Action: REJECT.

The 4:1s Rule

4

Consecutive

Four consecutive values.

1

Limit

Exceed ±1 SD.

→

Systematic

Must be on the same side.

Indicates a shift.

The 10x Rule

Definition

Ten consecutive values fall on the same side of the mean.

Error Type

Systematic Bias (Confirmed).

Even if values are within limits, this indicates a calibration issue or reagent problem.

Summary of Westgard Rules

Rule	Description	Error Type	Action
1:3s	1 point > ±3SD	Random	**REJECT**
2:2s	2 consec > ±2SD same side	Systematic	**WARNING**
R:4s	Range > 4SD	Random	**REJECT**
4:1s	4 consec > ±1SD same side	Systematic	**WARNING**
10x	10 on same side of mean	Systematic	**REJECT**

Applying Multiple Rules

A typical lab strategy processes rules in a specific order to save time:

```
1. Check 1:3s (Instant Rejection)
↓
2. Check 2:2s (Systematic Warning)
↓
3. Check 4:1s, 10x (Trend Detection)
```

Highlighting Violations in ggplot2

```
# Add violation column qc_data$Violation <- abs(qc_data$Value - mean) > 3*sd # Plot with conditional coloring ggplot(qc_data, aes(x = Day, y = Value)) + geom_line(color = "steelblue") + # Conditional color based on violation status geom_point(aes(color = Violation), size = 3) + scale_color_manual(values = c("FALSE" = "steelblue", "TRUE" = "red"))
```

Detecting Systematic Shifts

Signs of a Shift

- Values consistently on one side of mean.
- Sudden jump to a new level.
- 2:2s or 10x rules triggered.

Common Causes

- Reagent lot change (most common).
- Calibration drift.
- Maintenance performed on instrument.
- Environmental change (temp/humidity).

Visualizing a Shift

```r
# Mark shift point with vertical line
geom_vline(xintercept = 30.5,
           color = "purple",
           linetype = "dashed",
           linewidth = 1) +

# Add annotation
annotate("text", x = 40, y = upper_3sd,
         label = "SHIFT!", color = "red",
         fontface = "bold", size = 5)
```

Adding annotations helps tell the story of the data.

Trends vs. Shifts

Pattern	Description	Typical Cause
Shift	Sudden, abrupt jump to a new average level.	Reagent Lot Change, Lamp replacement.
Trend	Gradual movement up or down over time.	Reagent deterioration, Lamp aging, Debris buildup.

Complete QC Assessment Function

```
assess_qc <- function(values, mean, sd) { results <- data.frame( Point = 1:length(values), Value = values, Rule_1_3s = FALSE ) # Check 1:3s for all points results$Rule_1_3s <- abs(values - mean) > 3*sd # Add logic for other rules here... return(results) }
```

QC Decision Workflow

```
1. Run QC & Plot on LJ Chart

2. Check Westgard Rules

Any Violation?
YES → INVESTIGATE (Cause -> Correct -> Re-run)
NO → Release Results
```

Documenting QC Issues

CAP/CLIA Compliance Requirement: If you don't document it, it didn't happen.

What?

Rule triggered, Value, Lot #

Why?

Investigation findings (e.g., "Bubble in tubing")

Action?

Corrective action (e.g., "Primed lines")

Who?

Date, Time, Initials

Key Takeaways

- ✓ **LJ Charts** visualize QC performance over time.
- ✓ **Baseline** statistics (Mean/SD) define your limits.
- ✓ **Westgard Rules** (1:3s, 2:2s, etc.) detect specific error types.
- ✓ **Shifts** indicate systematic issues like reagent changes.
- ✓ **Documentation** is as important as the testing itself.

Tutorial 8: The QC Dashboard

Let's build your own QC monitoring tool.

Open your R environment

Tutorial 8: The QC Dashboard

Building a Levey-Jennings Chart in R

Your Scenario

The Role

You are the **QC Officer**.

The Data

60 days of Creatinine QC data.

The Problem

Around **Day 31**, something changed.

Goal: Visualize, Detect, and Report.

What You'll Do

Step	Task	Time
1	Load data	2 min
2	Calculate baseline (Days 1-20)	5 min
3	Calculate control limits	5 min
4	Create LJ chart	10 min
5	Flag & Highlight violations	10 min

Step 2: Calculate Baseline

Remember to only use the first 20 days!

```
baseline <- qc_data$Creatinine[1:20]

baseline_mean <- mean(baseline)
baseline_sd <- sd(baseline)

# Expected: Mean ~ 1.00, SD ~ 0.03
```

Step 4: LJ Chart

Key elements to include:

- Mean line (Green)
- ±2SD lines (Orange, Dashed)
- ±3SD lines (Red, Solid)
- Data points and connecting line

```
geom_hline(yintercept = upper_3sd, color = "red")
```

Step 5: Flag Violations

Implement the **1:3s**

Rule:

```
qc_data$Violation <- abs(qc_data$Creatinine - baseline_mean) > 3 * baseline_sd
```

Then count them using sum().

Step 6: Highlighting

Use conditional data subsets in ggplot to change point shapes and colors.

```
geom_point(data = qc_data[qc_data$Violation, ],
           size = 3, color = "red", shape = 17)
```

Shape 17 is a filled triangle▲.

What Your Chart Should Show

Days 1-30

Values oscillate randomly around the mean (~1.00).

Days 31-60

Systematic **Shift UP**.

Multiple points breaking the +3SD limit.

Clear visual separation.

Final Interpretation

Questions to Answer:

1. When did the shift occur?
2. What likely caused it?
3. What action would you take?

Typical Causes

Reagent Lot Change

Calibration Drift

Instrument Maintenance

Session 8 Complete

Great job building your QC Dashboard!